Pre-calculus 2

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Lecture 1: Sets and Numbers

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Definition 1 (Set). A set is a collection of objects specified in a manner that enables one to determine if a given object is or isn't in the set.

Exercise 1. Which of the following represent a set?

1 The students registered for MTH 112 at PCC this quarter.

2 The good students registered for MTH 112 at PCC this quarter.

Notation. Roster Notation involves listing the elements in a set within curly brackets: "{} "

Definition 2 (Element). An object in a set is called an **element** of the set. (symbol: "$\in$")

Example. 5 is an element of the set $\{4, 5, 6, 7, 8, 9\}$. We can express this symbolically:

$$5 \in \{4, 5, 6, 7, 8, 9\}$$

.

Definition 3 (Subset). A set S of a set T, denoted $S \subseteq T$, if all elements of S are also elements of T.

If S and T are sets and $S = T$, then $S \subseteq T$. Sometimes it's useful to consider a subset S of a set T that isn't equal to T. In such case, we write $S \subset T$ and say that S is a proper subset of T.

Example. $\{4, 7, 8\}$ is a subset of the set $\{4, 5, 6, 7, 8, 9\}$.

We can express this fact symbolically by $\{4, 7, 8\} \subseteq \{4, 5, 6, 7, 8, 9\}$

Since these two sets aren't equal, $\{4, 7, 8\}$ is a proper subset of $\{4, 5, 6, 7, 8, 9\}$, so can write:

$$\{4, 7, 8\} \subset \{4, 5, 6, 7, 8, 9\}$$

.

Definition 4 (Empty Set). The empty set, denoted $\emptyset$, is the set with no elements

$$\emptyset = \{\}$$

.

Definition 5 (Union)**.** The union of two sets A and B, denoted $A \cup B$, is the set containing all of the elements in either A or B (or both A and B).

Example. Consider the sets $\{4, 7, 8\}$, $\{0, 2, 4, 6, 8\}$, and $\{1, 3, 5, 7\}$. Then ...

- $\{4, 7, 8\} \cup \{1, 3, 5, 7\} = \{1, 3, 4, 5, 6, 8\}$
- $\{4, 7, 8\} \cup \{0, 2, 4, 6, 8\} = \{0, 2, 4, 6, 7, 8\}$
- $\{0, 2, 4, 6, 8\} \cup \{1, 3, 5, 7\} = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$

Definition 6 (Intersection)**.** The intersection of two sets A and B, denoted $A \cap B$, is the set containing all the elements in both A and B.

Example. Consider the sets $\{4, 7, 8\}$, $\{0, 2, 4, 6, 8\}$, and $\{1, 3, 5, 7\}$. Then ...

- $\{4, 7, 8\} \cap \{0, 2, 4, 6, 8\} = \{4, 8\}$
- $\{4, 7, 8\} \cap \{1, 3, 5, 7\} = \{7\}$
- $\{0, 2, 4, 6, 8\} \cap \{1, 3, 5, 7\} = \emptyset$

All the whole numbers form a set. This set is called the integers, and is represented by the symbol $\mathbb{Z}$. We can express the set of integers in roster notation:

$$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$$

.

Now, we can use set builder notation to create a set with a range of numbers.

Notation. Set Builder Notation.
"All the whole numbers between 3 and 10" $= \{x | x \in \mathbb{Z} \text{ and } 3 < x < 10\}$

Definition 7 (Important Sets of Numbers)**.** The set of natural numbers:

$$\mathbb{N} = \{1, 2, 3, 4, 5, \ldots\}$$

.

The set of integers:

$$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$$

.

The set of rational numbers:

$$\mathbb{Q} = \left\{x | x = \frac{p}{q} \text{ and } p, q \in \mathbb{Z} \text{ and } q \neq 0\right\}$$

.

The set of real numbers: $\mathbb{R}$

(All the numbers on the number line)

.

The set of complex numbers:

$$\mathbb{C} = \left\{x | x = a + bi \text{ and } a, b \in \mathbb{R} \text{ and } i = \sqrt{-1}\right\}$$

.

Note. $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$, the set of natural numbers ($\mathbb{N}$) is a subset of the set of integers ($\mathbb{Z}$) which is a subset of the set of rational numbers ($\mathbb{Q}$) which is a subset of the set of real numbers ($\mathbb{R}$) which is a subset of the set of complex numbers ($\mathbb{C}$).

Notation. Since we use the real numbers so often, we have special notation for subsets of the real numbers. **Interval Notation**. Interval Notation involves square or round brackets.

Example. Quick demo of Interval Notation

- $\{x | x \in \mathbb{R} \text{ and } -2 \leq x \leq 3\} = [-2, 3]$
- $\{x | x \in \mathbb{R} \text{ and } -2 < x < 3\} = (-2, 3)$
- $\{x | x \in \mathbb{R} \text{ and } -2 < x \leq 3\} = (-2, 3]$
- $\{x | x \in \mathbb{R} \text{ and } -2 \leq x < 3\} = [-2, 3)$

When the interval has no upper or lower bound, we use the infinity symbol (∞ or $-\infty$)

- $\{x | x \in \mathbb{R} \text{ and } x \leq 4\} = (-\infty, 4]$
- $\{x | x \in \mathbb{R} \text{ and } x \geq 4\} = [4, -\infty)$

Exercise 2. Simplify the following expressions:

- $(-4, \infty) \cup [-8, 3]$
- $(-4, \infty) \cup (-\infty, 2]$
- $(-4, \infty) \cap (-\infty, 2]$
- $(-4, \infty) \cap [-10, -5]$

Lecture 2: Angles and Arc-Length

Apr 06 2022 (07:25:34)

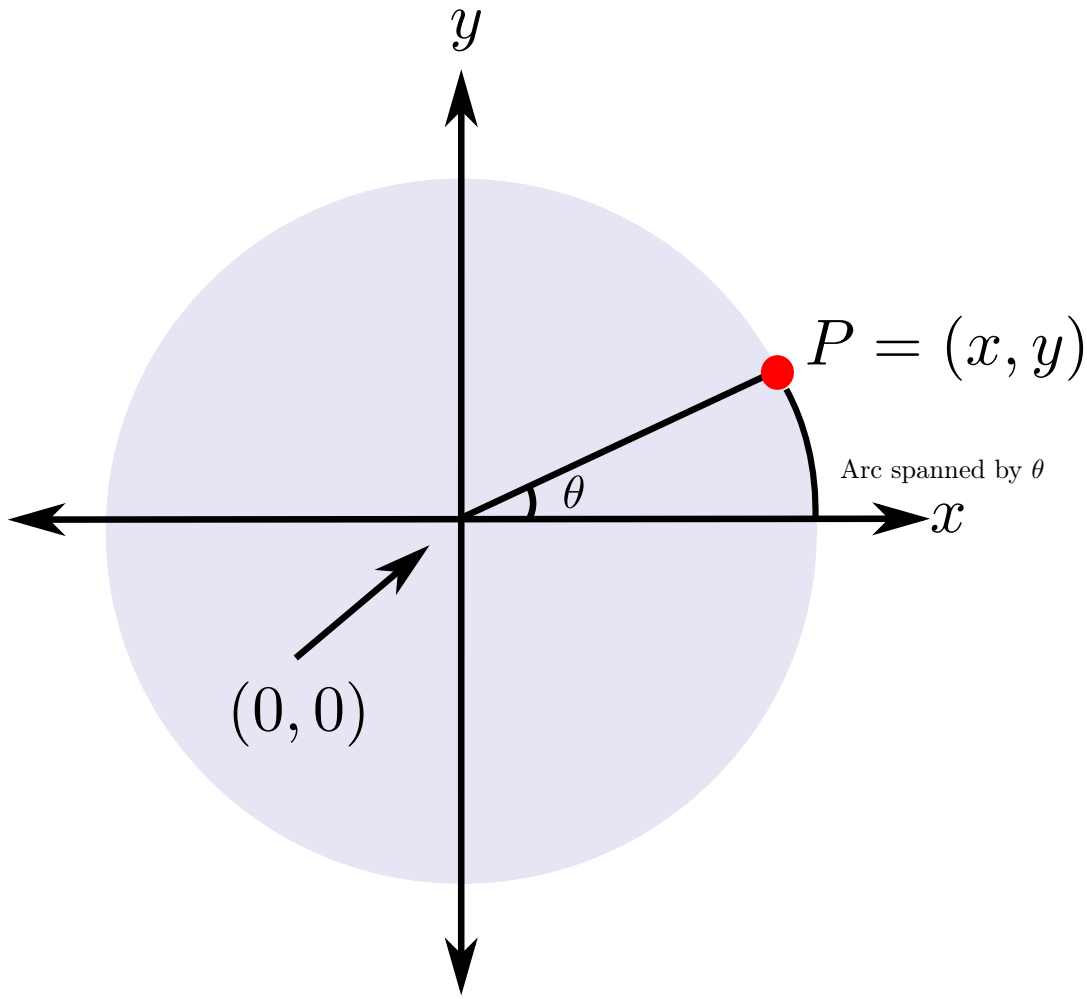

Figure 1

- The standard way that we put an angle in a circle we start the angle at the positive x-axis and go **counterclockwise**. We measure negative angles **clockwise**.
- When put an angle in standard position and you rotate, it ends someplace. Where it ends is called the **Terminal Side**.
- The point P on the circumference of the circle is **specified by the angle** θ.

- Angle θ corresponds with a portion of the circumference of the circle called the **arc spanned by** θ
- Two angles with the same terminal side are called **co-terminal angles**.

Three hundred and sixty degrees ($360°$) represents a complete trip around a circle, which is a full rotation. So, $1°$ corresponds to $\frac{1}{360°}$ of a full rotation. Degrees are more like percentages, where they represent concepts, and not numbers. But, just like how you can transform a percentage to a number ($10\% = \frac{10}{100}$), you can do the same with degrees ($10° = \frac{10}{360}$).

Since $360°$ represents a full rotation around the circle, if we add any integer multiple of $360°$ to an angle θ_1, we'll obtain an angle co-terminal to θ_1.

Example. So $45°$ and $405°$ are co-terminal.

Definition 8 (π)**.** The number π represents the ratio of the circumference of a circle to the diameter of the circle. $\pi = \frac{c}{d}$, where c is the circumference and d is the diameter.

$\pi \approx 3.14$, but π is an **"irrational"** number.

Definition 9 (Radian)**.** The **radian** measure of an angle is the ratio of the length of the arc on the circumference of the circle spanned by the angle, s, and the radius, r, of the circle.

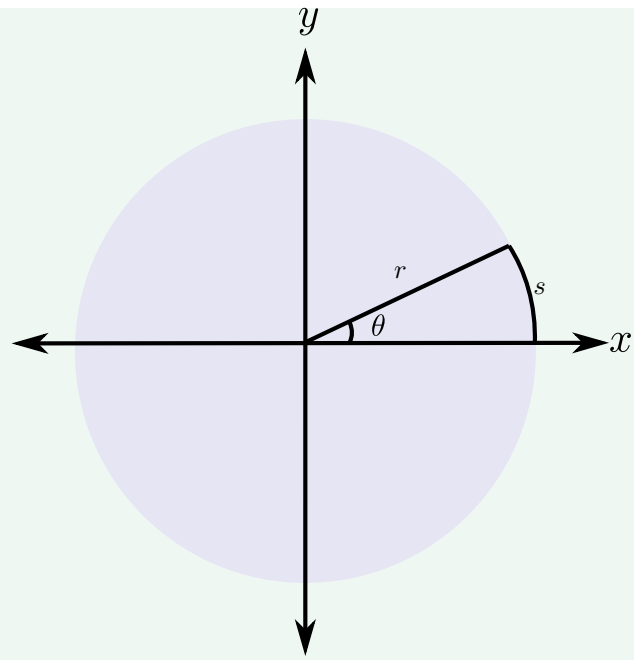

Figure 2: θ measures $\frac{s}{r}$ radian.

$$\begin{aligned} \pi &= \frac{c}{d} \\ &= \frac{2}{2r} \qquad d = 2r \qquad . \\ 2\pi &= \frac{c}{r} \qquad \text{Multiply both sides by 2} \end{aligned} \tag{1}$$

From this, we can conclude that: $2\pi = 360°$. From that, we can get $\pi = 180°$ since we just divided both sides by 2. From that, we get $\frac{\pi}{2} = 90°$. So on, and so forth.

Here's a table with all of the radians:

θ (degrees)	$0°$	$30°$	$45°$	$60°$	$90°$	$180°$	$270°$	$360°$
θ (radians)	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π

Table 1: Degrees into Radians

Exercise 3. Convert 8 radians into degrees and 8 degrees into radians:

Definition 10 (Arc Length)**.** The **arc length** s spanned in a circle of radius

r by an angle θ measured in radians is given by:

$$s = r \times \mid \theta \mid$$

.

Note. This formula only applies if θ is measured in radians.

Exercise 4. What is the arc length spanned by a $40°$ angle on a circle radius 30 meters?

Lecture 3: Introduction to Periodic Functions

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Any activity that repeats on a regular time interval can be described as *periodic*.

Definition 11 (Periodic Function). A **periodic function** whose values repeat on regular intervals. Hence, f is a periodic if there exists some constant c such that:

$$f(x+c) = f(x)$$

.

for all x in the domain of f such that $f(x+c)$ is defined.

Recall that this means that if the graph $y = f(x)$ is shifted horizontally c units then it will appear unaffected.

Definition 12 (Period). The **period** of a function f is the smallest value $\mid c \mid$ such that $f(x+t)$ for all x in the domain of f such that $f(x+c)$ is defined.

Example. We know that it's a periodic since an interval of the graph repeats over and over and over.

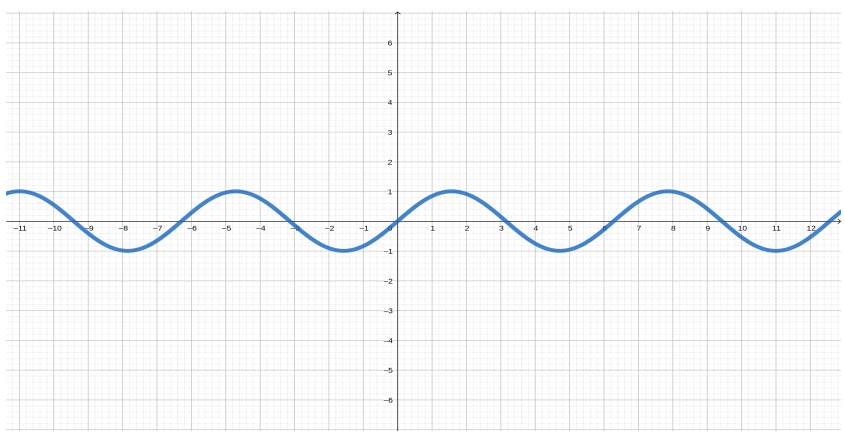

Figure 3: Periodic Function: $\sin(x)$

You can't really tell the period for this function, but the period for the function sin is 2π. One way to find the period of a graph is to find the distance from one minimum point to the other or one maximum point to another.

Definition 13 (Midline). The **midline** of a periodic function is the horizon-

tal line midway between the function's minimum and maximum values.

Lecture 4: Part 1: Intro to Trig Functions

Apr 12 2022 (16:27:32)

Definition 14 (Unit Circle). A **unit circle** is a circle with a radius of 1 unit.

Now we can define the sin and cos functions.

Note. There are 4 other trigonometric functions that can be defined in terms of sin and cos functions so first we'll get familiar with sin and cos and then later in this lecture we'll define the 4 other trigonometric functions.

Definition 15 (Sine and Cosine). The **sine function**, denoted $\sin(\theta)$, associates each angle θ with the vertical coordinate of the point P specified by θ on the circumference of a unit circle.

The **cosine function**, denoted $\cos(\theta)$, associates each angle θ with the horizontal coordinate of the point P specified by θ on the circumference of a unit circle.

So the point P in Figure 4 has coordinates:

$$(x, y) = (\cos(\theta), \sin(\theta))$$

.

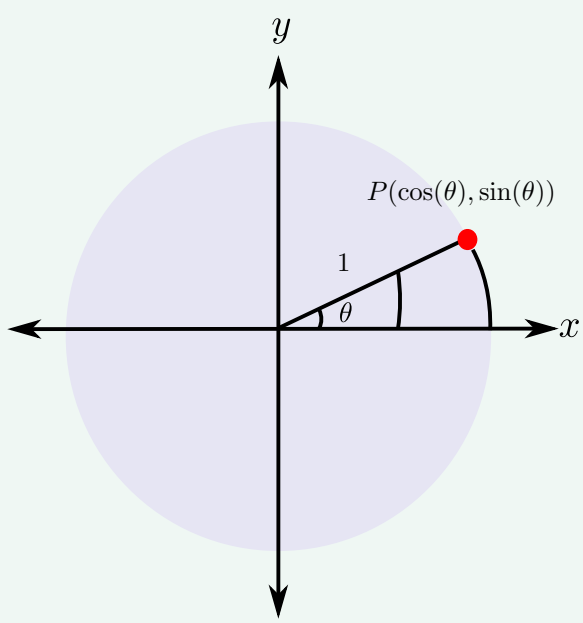

Figure 4

Exercise 5. The angle of θ specifies the point $P = \left(-\frac{3}{5}, \frac{4}{5}\right)$ on the circumference of a unit circle. Find $\sin(\theta)$ and $\cos(\theta)$.

Let's determine the signs of the sin and cos functions in the four quadrants:

- When the terminal side of angle θ is in Quadrant I, both the x and y coordinates of point P are positive. Therefore, **if θ is in Quadrant I,** $\cos(\theta) > 0$ **and** $\sin(\theta) > 0$
- When the terminal side of angle θ is in Quadrant II, the y coordinate of point P is positive but the x coordinate is negative. Therefore, **if θ is in Quadrant II,** $\cos(\theta) < 0$ **and** $\sin(\theta) > 0$
- When the terminal side of angle θ is in Quadrant III, both the x and y coordinates of point P are negative. Therefore, **if θ is in Quadrant IIII,** $\cos(\theta) < 0$ **and** $\sin(\theta) < 0$
- When the terminal side of angle θ is in Quadrant IIII, the x coordinate of point P is positive but the y coordinate is negative. Therefore, **if θ is in Quadrant IIII,** $\cos(\theta) > 0$ **and** $\sin(\theta) < 0$

Now let's find the sin and cos of a few particular angles. The easiest points for us to find on the unit circle are points where the circumference of the circle intersects the coordinate axes. Let's start by finding the corresponding sin and cos values.

Note. Keep in mind that cos represents the x-coordinate and sin represents the y-coordinate.

- The angle $\theta = 90°$ ($\theta = \frac{\pi}{2}$ radians), specifies the point $(0, 1)$ on the circumference of a unit circle. Thus ...

$$\cos\left(\frac{\pi}{2}\right) = 0 \qquad \sin\left(\frac{\pi}{2}\right) = 1$$

.

- The angle $\theta = 180°$ ($\theta = \pi$ radians), specifies the point $(-1, 0)$ on circumference of a unit circle. Thus ...

$$\cos(\pi) = -1 \qquad \sin(\pi) = 0$$

.

- The angle $\theta = 270°$ ($\theta = \frac{3\pi}{2}$ radians), specifies the point $(0, -1)$ on the circumference of a unit circle. Thus ...

$$\cos\left(\frac{3\pi}{2}\right) = 0 \qquad \sin\left(\frac{3\pi}{2}\right) = -1$$

.

- The angle $\theta = 360°$ ($\theta = 2\pi$ radians), specifies the point $(1, 0)$ on the circumference of a unit circle. Thus ...

$$\cos(2\pi) = 1 \qquad \sin(2\pi) = 0$$

.

Notice that angles of measure 2π radians and 0 radians specify the same point: $(1, 0)$. Thus, the sin and cos values for 2π and 0 radians are the same:

$$\cos(2\pi) = \cos(0) = 1 \qquad \text{and} \qquad \sin(2\pi) = \sin(0) = 0$$

.

Since any angle θ and $\theta + 2\pi$ specify the same point on the unit circle, the sin and cos values of θ and $\theta + 2\pi$ are the same. Therefore, the period of the sin and cos function is 2π radians.

For all θ, $\sin(\theta) = \sin(\theta + 2\pi)$ and $\cos(\theta) = \cos(\theta + 2\pi)$ so the period of both $s(\theta) = \sin(\theta)$ and $c(\theta) = \cos(\theta)$ is 2π radians.

Now, we'll sketch graphs of the sin and cos functions.
We first start by organizing the function values in a table:

Table 2: Values of Sin and Cos

θ (degrees)	$0°$	$90°$	$180°$	$270°$	$360°$	$450°$	$540°$	$630°$	$720°$
θ (radians)	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π	$\frac{5\pi}{2}$	3π	$\frac{7\pi}{2}$	4π
$y = \cos(\theta)$	1	0	-1	0	1	0	-1	0	1
$y = \sin(\theta)$	0	1	0	-1	0	1	0	-1	0

Here's how it looks like when we graph:

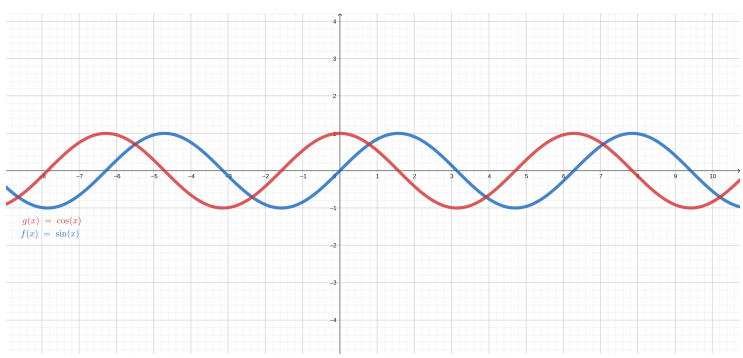

Figure 5: The graph of $y = \cos(\theta)$ and $y = \sin(\theta)$

Notice that the graphs of $y = \cos(\theta)$ and $y = \sin(\theta)$ are very similar. In fact, if we shift $y = \sin(\theta)$ to the left $\frac{\pi}{2}$ units, we'll obtain the graph of $y = \cos(\theta)$. This means that:

$$\cos = \sin\left(\theta + \frac{\pi}{2}\right)$$

.

Similarly, if we shift $y = \cos(\theta)$ to the right $\frac{\pi}{2}$ units, we'll obtain the graph of $y = \sin(\theta)$. This means that:

$$\sin = \cos\left(\theta - \frac{\pi}{2}\right)$$

.

Definition 16 (Identity)**.** An **identity** is an equation that is true for all values in the domains of the involved expressions.

Some important trig identities

- $\cos(\theta) = \cos(\theta + 2\pi)$
- $\sin(\theta) = \sin(\theta + 2\pi)$
- $\sin(\theta) = \cos\left(\theta - \frac{\pi}{2}\right)$
- $\cos(\theta) = \sin\left(\theta + \frac{\pi}{2}\right)$
- $\cos(-\theta) = \cos(\theta)$
- $\sin(-\theta) = -\sin(\theta)$
- $\cos(\theta) = \cos(2\pi - \theta)$
- $\sin(\theta) = \cos(\pi - \theta)$

We can generalize the definitions of sin and cos functions so that they are applicable to circles of any size, rather than only for unit circles.

Definition 17 (More Applicable version of Sin and Cos)**.** If the point $T = (x, y)$ is specified by the angle θ on the circumference of a circle of radius r then:

$$\cos(\theta) = \frac{x}{r} \qquad \text{and} \qquad \sin(\theta) = \frac{y}{r}$$

.

Note. If $r = 1$, then this definition $\cos(\theta)$ and $\sin(\theta)$ are equivalent to what we saw at the beginning of this lecture.

$$\cos(\theta) = \frac{x}{r} = \frac{x}{1} = x \qquad \text{and} \qquad \sin(\theta) = \frac{y}{r} = \frac{y}{1} = y$$

.

If we solve the equations $\cos(\theta) = \frac{x}{r}$ and $\sin(\theta) = \frac{y}{r}$ for x and y, we can obtain the coordinates of a point on the circumference of a circle of any r :

$$\cos(\theta) = \frac{x}{r} \Rightarrow x = r\cos(\theta) \qquad \text{and} \qquad \sin(\theta) = \frac{y}{r} \Rightarrow y = r\sin(\theta)$$

.

If the point $T = (x, y)$ is specified by the angle θ on the circumference of a circle of radius, r, then:

$$x = r\cos(\theta) \qquad \text{and} \qquad y = r\sin(\theta)$$

.

Solutions

Solution 1. for **Exercise** 1

1. This represents a set since it's well defined. We all know what it means to be registered for a class.
2. This does not represent a set since it's not well defined. There are many different interpretations of what it means to be a good student (get an **A** or pass the class or attend class or avoid falling asleep in class or don't cause trouble in class)

Solution 2. for **Exercise** 2

- $(-4, \infty) \cup [-8, 3] = [-8, \infty)$
- $(-4, \infty) \cup (-\infty, 2] = (-\infty, \infty) = \mathbb{R}$
- $(-4, \infty) \cap (-\infty, 2] = (-4, 2]$
- $(-4, \infty) \cap [-10, -5] = \emptyset$

Solution 3. for **Exercise** 3

Since $2\pi = 360°$ $\quad \frac{2\pi}{360°} = 1 = \frac{360°}{2\pi}$. We're trying to cancel out the radians. So, to do this, we will need to have the radians on the bottom, and the degrees on the top. After we do the multiplication, we will be left with the degrees.

$$\begin{aligned} 8\,\text{rad} \times \left(\frac{360°}{2\pi\,\text{rad}}\right) &= \frac{8 \times 360°}{2\pi} \\ &= \frac{1440°}{\pi} \\ &\approx 458.4. \end{aligned}$$

Now, let's convert 8° into 8 radians.

$$\begin{aligned} 8° \left(\frac{\pi\,\text{rad}}{180°}\right) &= \frac{8\pi}{180}\,\text{rad} \\ &= \frac{\pi}{45}\,\text{rad} \end{aligned}$$

.

Solution 4. for **Exercise** 4

Convert 40° into radians:

$$\begin{aligned} 40^\circ \times \left(\frac{\pi}{180^\circ}\right) &= \frac{40\pi}{180}\,\text{rad} \\ &= \frac{4\pi}{18} \\ &= \frac{2\pi}{9} \\ &\approx 0.7 \end{aligned}$$

.

Now we're ready to compute the arc length:

$$\begin{aligned} s &= r\times \mid \theta \mid \\ &= (30m) \times (\frac{2\pi}{9}) \\ &= \frac{20\pi}{3}m. \end{aligned}$$

Solution 5. for **Exercise** 5

$$\sin(\theta) = \frac{4}{5} \qquad \text{and} \qquad \cos(\theta) = -\frac{3}{5}$$

.